Documentation chemical data input

This document contains the documentation for the chemical data input in the program **"interfaz_remix"**. There are maximum 5 files to be read. The minimum of input files is 3 (the ones that end in *"_quim_loc.dat", "_sist_quim.dat"* and *"_tar_wat.dat"*). The other two (ending in *"_tar_sol.dat"* and *"_tar_gas.dat"*) are optional, depending if the chemical system contains solids and/or gases. All of the files must contain the **same root** in their names, and they must live in the **same directory**.

We describe the input files with an example of two waters that mix: an inflow water in equilibrium with atmospheric O_2 and CO_2 and a resident water in equilibrium with calcite. This example can be found in the directory *"..\examples\calcite_eq"* in the repository.

Chemical system file: *"_sist_quim.dat"*

This file contains the information related to the chemical system (species, reactions, etc). It is read by the **Chemical System** class.

- **Line 1**: Title of the problem
- **Line 2**: Section separation
- **Line 3**: Section name
    - If Section name = 'PRIMARY AQUEOUS SPECIES' or 'AQUEOUS COMPLEXES':
        1. **Line 3a**: *sp_name* (string)
            1. Name of species in database ***"master25_modif.dat"***
        2. **Line 3b**: end of section
    - If Section name = 'MINERALS' or 'GASES':
        1. **Line 3a**: *sp_name* (string), *flag_eq* (logical), *flag_cst_act* (logical)
            1. *sp_name:* Name of species in database ***"master25_modif.dat"***
            2. *flag_eq*: TRUE if equilibrium reaction, FALSE otherwise
            3. *flag_cst_act*: TRUE if species has constant activity, FALSE otherwise
        2. **Line 3b**: end of section
    - If Section name = 'REDOX REACTIONS':
        1. **Line 3a**: *react_name* (string), *flag_eq* (logical)
            1. *react_name:* Name of redox reaction in database ***"reacciones_monod.dat"***
            2. *flag_eq*: TRUE if equilibrium reaction, FALSE otherwise
        2. **Line 3b**: end of section
    - If Section name = 'SURFACE COMPLEXES':
        1. **Line 3a**: *sp_name* (string)
            1. *sp_name:* Name of species in database ***"master25_modif.dat"***
        2. **Line 3b**: end of section
- **Line 4**: Section separation
- **Line 5**: end of file

```
'TITLE OF THE PROBLEM: Calcite in equilibrium'
'---------------------------------------------------------------------------------------------------------------------------------------------------------------------------'
'PRIMARY AQUEOUS SPECIES'
'h2o(p)'                                        ! name of primary aqueous species
'h+'
'ca+2'
'o2(aq)'
'hco3-'
'*'
'MINERALS'
'calcite' .true. .true.                         ! name of mineral, flag equilibrium, flag constant activity
'*' .true. .true.
'GASES'
'o2(g)'  .true. .true.                          ! name of gas, flag equilibrium, flag constant activity
'co2(g)'  .true. .true.                         ! name of gas, flag equilibrium, flag constant activity
'*'  .true. .true.
'---------------------------------------------------------------------------------------------------------------------------------------------------------------------------'
'end'
```

Local chemistry file: *"_quim_loc.dat"*

This file contains the **local** chemical information (water types, mineral zones, gas zones). It is read by the **Chemistry** class.

- **Line 1**: Name of the problem
- **Line 2**: Section separation
- **Section name**: 'INITIAL AND BOUNDARY WATER TYPES'
  - **Line 1**: *act_coeffs_model* (integer)
    0. Ideal
    1. Debye-Hückel
    2. Debye-Hückel extended
    3. Davies
    4. Truesdell-Jones
  - **Line 2**: *num_wat_types* (integer)
    i. Number of water types (includes initial, recharge, boundary, etc)
  - **Line 3**: *ind_wat_type* (integer), *temp* (real)
    i. *ind_wat_type*: index of water type
    ii. *temp*: temperature of solution (in Celsius)
  - **Line 3a**: *name* (string)
    i. Name of water type
  - **Line 3b**: water type data begins
    i. <u>*icon*</u>: initial condition type
      1. prescribed primary concentration
      2. prescribed total concentration
      3. prescribed activity
      4. primary species is in equilibrium with a phase
    ii. <u>*guess*</u>: Initial guess for primary species concentration
    iii. <u>*ctot*</u>: (all concentrations are expressed in molalities)
      i. *icon=1*: primary species concentration
      ii. *icon=2*: total concentration of primary species
      iii. *icon=3*: activity of primary species
      iv. *icon=4*: activity of phase that is in equilibrium with this primary species
    iv. <u>*constrain*</u>: if *icon=4*, it is the phase in equilibrium with the primary species
  - **Line 3c:** *prim_sp_name* (string), *icon* (integer), *guess* (real), *ctot* (real), *constrain* (string)
  - **Line 3d**: end of water type data
  - **Line 4**: end of section

```
'TITLE OF THE PROBLEM: Calcite in equilibrium'
'-------------------------------------------------------------------------------------------------------------------------------------------------------------------------------'
'INITIAL AND BOUNDARY WATER TYPES'
0                                                   ! activity coefficients model
2                                                   ! number of water types
1    25.0                                           ! index water type, temperature (C)
'boundary'                                          ! name of water type
'     icon      guess       ctot        constrain'
'h2o(p)'    3         1d0         1d0                 ''
'ca+2'      1         1d-9        1d-9                ''
'o2(aq)'    4         1d-9        2d-1            'o2(g)'
'hco3-'     4         1d-9        3.5d-4          'co2(g)'
'h+'        1         1d-7        1d-7                ''
'*'     0       0.0       0.0         ' '
2    25.0                                           ! index water type, temperature (C)
'initial'                                           ! name of water type
'     icon      guess       ctot        constrain'
'h2o(p)'    3         1d0         1d0                 ''
'ca+2'      1         1d-3        1d-3                ''
'o2(aq)'    1         1d-6        1d-6                ''
'hco3-'     4         1d-9        1d0             'calcite'
'h+'        1         1d-5        1d-5                ''
'*'     0       0.0       0.0         ' '
'-------------------------------------------------------------------------------------------------------------------------------------------------------------------------------'
```

- Section name: 'INITIAL MINERAL ZONES'
  - **Line 1**: *num_min_zones* (integer)
    1. Number of mineral zones
  - **Line 2a**: *ind_min_zone* (integer), *temp* (real)
    1. *ind_min_zone*: index of mineral zone
    2. *temp*: temperature of mineral zone (in Celsius)
  - **Line 2b**: *name* (string)
    1. Name of mineral zone
  - **Line 2c**: mineral zone data begins
  - **Line 2d**: *min _name* (string), *vol_frac* (real), *react_surf* (real)
    1. *min_name*: name of mineral in mineral zone (must be in the chemical system)
    2. *vol_frac.:* initial volumetric fraction of mineral
    3. *react_surf*: specific surface of mineral (surface of mineral per unit volume of rock)
  - **Line 2e**: end of mineral zone data
  - **Line 3**: end of section

```
'INITIAL MINERAL ZONES'

1                                                        ! number of mineral zones

1        25.0                                            ! index of mineral zone, temp (C)

'calcite'                                                ! name of mineral zone

'mineral      vol.frac.    area'

'calcite'      0.50       0.1d+3

'*'         0.00       0.00

'-----------------------------------------------------------------------------------------------------------------------------------------------'
```

- Section name: 'INITIAL AND BOUNDARY GAS ZONES'
  - **Line 1**: *num_gas_zones* (integer)
    1. Number of gas zones
  - **Line 2a**: *ind_gas_zone* (integer), *temp* (real), *vol* (real)
    1. *ind_gas_zone*: index of gas zone
    2. *temp*: temperature of gas zone (in Celsius)
    3. *vol*: total volume of gas
  - **Line 2b**: gas zone data begins
  - **Line 2c**: *name* (string)
    1. Name of gas zone
  - **Line 2d**: *gas _name* (string), *part_press* (real)
    1. *gas_name*: name of gas in gas zone (must be in the chemical system)
    2. *part_press:* initial partial pressure (equivalent to activity) of gas (in atm)
  - **Line 2e**: end of gas zone data
  - **Line 3**: end of section
  - **Line 4**: end of file

```
'INITIAL AND BOUNDARY GAS ZONES'

1                                        ! number of gas zones

1     25     1d9                         ! index gas zone, temp (ºC), volume

'atmosphere'                             ! name of gas zone

'gas               partial pressure'

'o2(g)'                 2d-1

'co2(g)'                3.5d-4

'*'                     0.00

'------------------------------------------------------------------------------------------'

'end'
```

Target waters file: *"_tar_wat.dat"*

This file contains the **association** between target waters, target solids and/or target gases, and it also indicates if the target waters are external or not. This file is read by the **Chemistry** class.

- **Line 1**: Name of the problem
- **Line 2**: Section separation
- **Section name**: 'TARGET WATERS'
  - **Line 1**: *num_tar_wat* (integer)
    - Number of target waters
  - **Line 2**: *num_rech_wat* (integer)
    - Number of recharge target waters
  - **Line 3**: *num_bd_wat* (integer)
    - Number of boundary target waters
  - **Lines 4 - 4+nº intervals**: *int_wat* (string), *iwtype* (integer), *int_sol* (integer), *int_gas(*integer), *flag_wat_type*(logical)
    - *int_wat*: Interval of target waters
    - *iwtype*: Index water type in section 'INITIAL AND BOUNDARY WATER TYPES'
    - *int_sol*: Interval of target solids associated with *int_wat*
    - *int_gas*: Interval of target gases associated with *int_wat*
    - *flag_wat_type:* Flag to indicate target water type.
      - 0: Boundary
      - 1: Domain
      - 2: Recharge
  - End of section

```
'TITLE OF THE PROBLEM: Calcite in equilibrium'

'------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------'

'TARGET WATERS'

11                                              ! number of target waters

0                                               ! number of recharge target waters

1                                               ! number of boundary target waters

1        1        0        1        0           ! target waters interval, water type index, target solids interval, target gases interval, flag water type

2-11     2        1-10     0        1

'------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------'

'end'
```

Target solids file: *"_tar_sol.dat"*

This file contains the **association** between target solids and solid zones. It is read by the **Chemistry** class. The only solid zones implemented so far are the **mineral** zones. **IMPORTANT**: this file will not be read by the code if there are no solid zones.

- **Line 1**: Name of the problem
- **Line 2**: Section separation
- **Section name**: 'TARGET SOLIDS'
  - **Line 1**: *num_tar_sol* (integer)
    - Number of target solids
  - **Line 2**: *interval* (string), *iszone* (integer)
    - *interval*: Target solids interval
    - *iszone:* Index of mineral zone in section 'INITIAL MINERAL ZONES'
  - end of section

```
'TITLE OF THE PROBLEM: Calcite in equilibrium'

'------------------------------------------------------------------------------------------------------------------------------------------'

'TARGET SOLIDS'

10                                          ! number of target solids

1-10        1                               ! target solids interval, solid zone index

'------------------------------------------------------------------------------------------------------------------------------------------'

'end'
```

Target gases file: *"_tar_gas.dat"*

This file contains the **association** between target gases and gas zones. It is read by the **Chemistry** class. **IMPORTANT**: this file will not be read by the code if there are no gas zones.

- **Line 1**: Name of the problem
- **Line 2**: Section separation
- **Section name**: 'TARGET GASES'
  - **Line 1**: *num_tar_gas* (integer)
    - Number of target gases
  - **Line 2**: *interval* (string), *igzone* (integer)
    - *interval*: Target solids interval
    - *igzone:* Index of gas zone in section 'INITIAL AND BOUNDARY GAS ZONES'
  - end of section

```
'TITLE OF THE PROBLEM: Calcite in equilibrium'

'------------------------------------------------------------------------------------------------------------------------------------------'

'TARGET GASES'

1                                           ! number of target gases

1           1                               ! target gases interval, gas zone index

'------------------------------------------------------------------------------------------------------------------------------------------'

'end'
```